of paired nitrogenous bases, adenine (A), guanine (G), cytosine (C), and thymine (T). A always pairs up with T to form base pair AT, and G always pairs with C to form base pair GC.

DNA is the blueprint for life. Sequences of bases along the DNA strand constitute the genes that provide the information to determine the complete form and physiology of an organism. A triplet of bases is known as a codon, and each codon specifies the production of one of 20 amino acids; the order in which the amino acids join together in a chain determines the type of protein they go on to make. For example, the combination GGA is the codon for glycine. Sixty-four possible triplets can be made from the four base pairs, and 61 of them code for a particular amino acid. The other three act as signals such as “start” and ”stop,” which govern how information is read by the cellular machinery. DNA is also organized into separate chromosomes, of which there are 23 pairs in the human cell.